ModelIt!

Student Activity Pack

How Do Noise-Canceling Headphones Work?

Modeling Wave Superposition and Destructive Interference

Grade 11L1 | NGSS HS-PS4-1, HS-PS4-5 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How do you think noise-canceling headphones work — do they block sound, absorb it, or do something else entirely?**

2. **What happens when two waves meet in the same place at the same time? Can you think of examples from water or sound?**

3. **Why do you think noise-canceling headphones are better at canceling airplane engine noise than they are at canceling human voices?**

4. **If I told you that noise-canceling headphones work by adding MORE sound to your ears, would you believe me? How could more sound make things quieter?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Ambient Noise Frequency | Ambient Noise Amplitude | Processing Speed | Phase Accuracy | Cancellation Effectiveness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Low-Frequency Cancellation

Noise: 100 Hz sine wave at 80 dB | ANC: Active

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High-Frequency Challenge

Noise: 4,000 Hz sine wave at 70 dB | ANC: Active

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Mixed Environment

Noise: Airplane cabin spectrum (multiple frequencies) | ANC: Active

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Feedback vs. Feedforward Architecture — Two approaches to ANC microphone placement: feedforward places the mic outside the ear cup (sensing noise before it arrives) while feedback places it inside (sensing the combined result). Each has trade-offs in latency, accuracy, and frequency range

Ear Seal Quality — The physical fit and acoustic seal between the headphone ear cup and the wearer's head — passive isolation from the physical seal blocks 10-25 dB of high-frequency noise that active cancellation struggles with, making passive and active noise reduction complementary

Adaptive Algorithm — Machine learning algorithms that learn the acoustic characteristics of common noise environments (airplane, train, office) and pre-load optimized anti-phase profiles, reducing the processing time needed for real-time cancellation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can adding more sound to your ears actually make everything quieter — and what does this tell us about the hidden mathematical nature of waves? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that noise cancellation is based on wave mathematics (superposition) rather than physical sound blocking?

2 Why does the same timing error that barely affects low-frequency cancellation completely ruin high-frequency cancellation?

3 What does the frequency-dependent effectiveness of noise cancellation reveal about the fundamental relationship between wavelength and precision?

4 How could the principles of destructive interference be applied beyond headphones — in architecture, vehicles, or industrial noise control?

5 What limitations does your model have in capturing the full complexity of real-world noise cancellation environments?

8 STEM Challenge: Design a Next-Generation Noise Cancellation System

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YOUR ENGINEERING MISSION

Design an improved active noise cancellation system that extends effective cancellation to higher frequency ranges than current technology allows.

THE SCENARIO

A headphone manufacturer wants to develop the next generation of noise-canceling headphones that can cancel not just engine rumble but also human speech and other mid-frequency noise. Current technology achieves 25 dB reduction below 500 Hz but only 5 dB reduction above 2,000 Hz. Your acoustic engineering team must design a system architecture that improves high-frequency cancellation using faster processing, predictive algorithms, or multi-microphone arrays.

Engineering Design Thinking

1. WHAT MICROPHONE PLACEMENT STRATEGY WOULD MINIMIZE THE DISTANCE SOUND MUST TRAVEL BEFORE BEING SAMPLED, REDUCING PROCESSING DELAY?

2. HOW COULD MULTIPLE MICROPHONES CREATE A SPATIAL NOISE MAP THAT PREDICTS INCOMING SOUND DIRECTION AND CHARACTERISTICS?

3. WHAT ROLE COULD MACHINE LEARNING PLAY IN PRE-COMPUTING ANTI-PHASE SIGNALS FOR COMMON NOISE ENVIRONMENTS (AIRPLANE, SUBWAY, OFFICE)?

4. HOW WOULD YOU TEST YOUR IMPROVED SYSTEM, AND WHAT METRICS WOULD PROVE IT OUTPERFORMS CURRENT TECHNOLOGY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Acoustic Engineers design sound management systems for headphones, concert venues, automotive cabins, and architectural spaces, earning $70,000–$130,000/year. Audio DSP (Digital Signal Processing) Engineers develop the real-time algorithms that compute anti-phase signals in noise-canceling headphones, earning $90,000–$170,000/year at companies like Bose, Sony, and Apple. Acoustics Researchers study wave behavior and develop new approaches to sound control, earning $75,000–$140,000/year.